/ winter 2020

CN 3150710

Protocol

★ Ethernet
↳ used LAN technology
→ 1. switched ethernet
→ 2. Fast ethernet
→ 3. Gigabyte ethernet

★ CSMA (Carrier Sense Multiple Access)
→ 1. 1- persistent CSMA
→ 2. Non Pessistent CSMA
→ 3. P- persistent CSMA
→ 4. CSMA/CD (CSMA with Collision Detection)

CRC is avery effective detection technique.

★ CRC: Cyclic Redundency Check

★ The count of Infinity problem

→ Vector problem

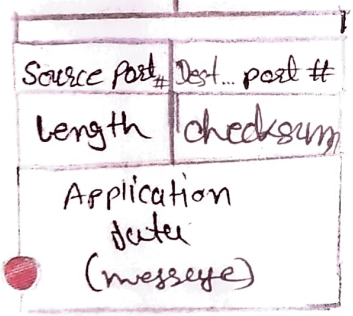
A	B	C	D	E	
•	•	•	•	•	initially
1	•	•	•	•	after 1 Exchange
1	2	•	•	•	after 2 exchange
1	2	3	•	•	after 3 exchange
1	2	3	4	•	after 4 exchange

A	B	C	D	E	
1	2	3	4		initially
3	2	3	4		after 1 exchange
3	4	3	4		after 2 exchange
5	4	5	4		after 3 exchange
5	6	5	6		after 4 exchange
7	6	7	6		after 5 exchange
7	8	7	8		after 6 exchange
.	.	.	.		

★ $d_x(y) = \min_v\{c(x,v) + d_v(y)\}$ → DVR

★ Supernetting:
→ Supernetting or CIDR (classless Interdomain Routing) is a way to aggregate multiple Internet addresses of the same class.

★ UDP Segment Structure



★ TCP segment structure

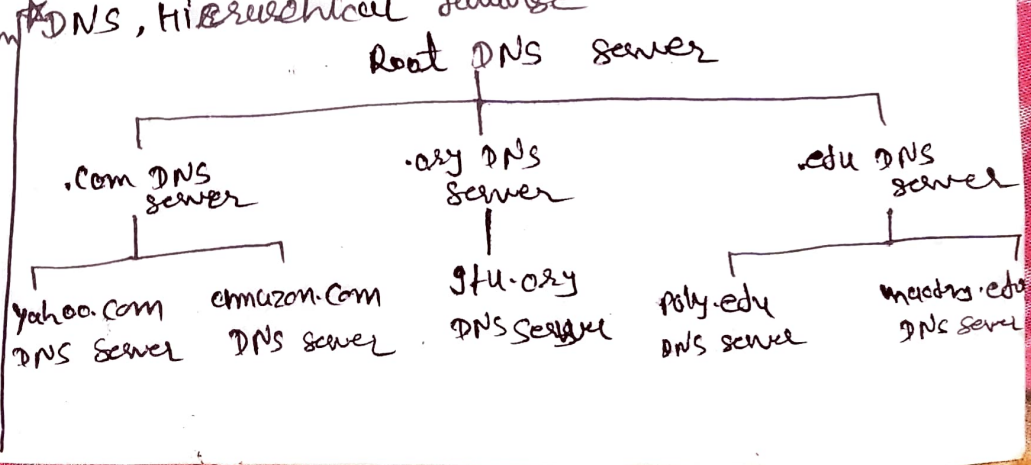
Source Port #	Destination port #
sequence number	
Acknowledgement number	
Header Length | Unused | URG | ACK | PSH | RST | SYN | FIN	Receiver window
Internet checksum	Urgent data pointer
options	
Data	

IMAP → 143
SSH → 22
DHCP → 67, 68
DNS → 53

★ Lecky Bucket Algorithm

★ Port Number
FTP → 20, 21
TELNET → 23
HTTP → 80, 8080
HTTPS → 443
SMTP → 161
POP3 → 110

★ DNS, Hierarchical database



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★Topology: Network topology is the schematic description of a network arrangement, connecting various nodes (sender and receive) through lines of connection.

→1) Bus topology 3) Star topology 5) Tree topology
2) Ring 4) Mesh 6) Hybrid

★Delay: As a packet travel from one node (host or router) to the subsequent node along this path, the packet suffer from several types of delays at each node along the path.
→1. Nodal processing Delay →3. Transmission Delay
→2. Queuing Delay →4. Propagation delay

HUB, Switch

★1) Repeater: Repeater is an electronic device. It operates only the physical layer of the OSI model. Repeater generates the weak signal and extend the limit of the network.

→2) Bridge: It operates at the data link layer. I connects two network together.

→3) Router: It connects more than two different types of network. Router determine the short path between source and destination for the data transmission.

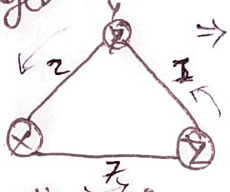
→4) Gateway: It operates at the application layer of OSI model. It connects two dissimilar networks together.

1. Persistent
2. Non-Persistent
↑types

★HTTP ★DNS ★OSI Model ★DHCP ★Email ★ARP
★Dynamic Host configuration Protocol. ★Address Resolution Protocol

★Vector Routing Algorithm (DV), Example → $d_x(y) = \min_v\{c(x,v) + d_v(y)\}$

→iterative
→asynchronous
→distributed



⇒ Node X table

cost to	X	Y	Z
X	0	2	7
Y	∝	∝	∝
Z	∝	∝	∝

from

	X	Y	Z
X	0	2	3
Y	2	0	3
Z	3	1	0

	X	Y	Z
X	0	2	3
Y	2	0	1
Z	7	1	0

★ARQ
★SNMP ★RIP ★VoIP
→Simple Network Management Protocol
→Routing Information Protocol
→Voice Over Internet Protocol
★RSVP
→Resource reserVation Protocol
★POP3 ★IMAP
→Post office Protocol
→Internet Message Control Protocol.
★Automatic Repeat Request.

⇒ Node Y table

	X	Y	Z
X	∝	∝	∝
Y	2	0	1
Z	∝	∝	∝

	X	Y	Z
X	0	2	7
Y	2	0	1
Z	7	1	0

	X	Y	Z
X	0	2	3
Y	2	0	1
Z	3	1	0

⇒ Node Z table

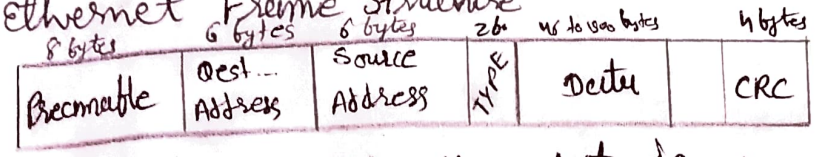
	X	Y	Z
X	∝	∝	∝
Y	∝	∝	∝
Z	7	1	0

	X	Y	Z
X	0	2	7
Y	2	0	1
Z	3	1	0

	X	Y	Z
X	0	2	3
Y	2	0	1
Z	3	1	0

☆ Example of Random Access Protocol (Medium Access Control)
→1. ALOHA (Advocates of Linux Opensource Hawaii Association)
→2. Slotted ALOHA
→3. CSMA, CSMA/CD, CSMA/CA

☆ Ethernet frame Structure



☆ Unicasting: In this the data is transffered from single sender (single source) to a single receiver (single destination)

☆ Broadcasting: In this, the data is transmitted from one or more network sender to all the receivers within the same network or in other

☆ Multicasting: When a data is transmitted from a single source to a specific group of hosts having the interest to receive the data, It is known as Multicast.

☆ MIME: Multipurpose Internet Mail Extension

☆ HUB: A hub is basically multiport repeater. A hub connects multiple wires coming from different branches, i.e: star topology.

☆ Bridge switch: A switch is a multiport bridge with a buffer and design that can boost it's efficiency and performance.

☆ Types of Switching Fabrics
↳1. Switching via memory
↳2. Switching via Bus
↳3. switching via interconnection network

☆ socket
☆ Multiplexing
☆ Demultiplexing

☆ Connection oriented v/s connectionless service (TCP, UDP, SCTP)

☆ Intra-AS Routing: It is also known as interior gateway protocol (IGP)
Types ↳1. RIP: Routing Information Protocol
↳2. OSPF: Open Shortest Path First
↳3. IGRP: Interior Gateway Routing protocol.

☆ Reliable Data Transfer
↳1. rdt1.0: rdt over a Perfectly Reliable channel.
↳2. rdt2.0: RDT over a channel with bit errors.
↳3. rdt2.1: sender handles garbled ACK/NAKs